

The impact of climate policy uncertainty on corporate green innovation: Evidence from China

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July 24, 2025

Abstract

This paper examines the impact of climate policy uncertainty (CPU) on corporate green innovation via a two-way fixed effects model and instrumental variable (IV) method, utilizing panel data from Chinese A-share listed companies spanning 2010 to 2022. The findings reveal that CPU positively affect corporate green innovation. Further heterogeneity analyses reveal that this positive impact is more pronounced in highly polluting firms, non-state-owned enterprises, companies led by managers with stronger green awareness, firms with lower risk-taking tendencies, and those facing fewer financing constraints. This study offers critical insights and empirical evidence for understanding corporate green decision-making in the context of CPU.

Keywords: Corporate green innovation ; Enterprise decision-making; Climate policy uncertainty

1 Introduction

In the face of global climate change, governments worldwide have responded vigorously by introducing a series of climate policies aimed at controlling greenhouse gas emissions and contributing to mitigating climate change. Climate change policies are long-run in nature and are inherently uncertain. Climate policy uncertainty (CPU) refers to the unpredictability of the content and effects of climate policies (Addey and Nganje, 2024), a factor that influences market behavior and investment decisions.

China has introduced numerous policies to address climate change, yet macroeconomic fragility and international conflicts have led to unavoidable adjustments. Despite a large continuity in climate change policies, the CPU Index rose from 1.053 to 2.125 between 2000 and 2022 (Ma et al., 2023). Green innovation is vital for companies pursuing low-carbon transformation, aligning with sustainability goals while enhancing competitiveness (Yuan and Cao, 2022; Fernando and Wah, 2017). However, the substantial capital investment and risks associated with green innovation projects (Kong et al., 2022) make them highly susceptible to external policy uncertainties, underscoring the need for a deeper analysis of CPU's impact on corporate green innovation.

While extensive research explores the link between policy uncertainty and innovation, consensus remains elusive. According to real options theory, increased CPU hampers firms' ability to assess external risks, leading to more cautious investment decisions that may stifle green R&D and constrain green technology innovation (Bloom et al., 2007; Sun et al., 2024). Conversely, it is arguable that CPU can create

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strategic opportunities, enabling firms to leverage green technology to enhance their market influence and accelerate innovation (Bai et al., 2023; Zhu et al., 2023).

In this paper, we study the the impact of CPU on corporate green innovation in China by utilizing provincial-level CPU data in China to assess its impact on corporate green innovation, addressing the limitation of prior research that primarily relies on national-level data. We employ the provincial-level CPU index in China, as introduced by Ma et al. (2023), and integrate it with panel data from A-share listed firms. Our findings reveal that CPU significantly positively impacts corporate green innovation, and this conclusion remains robust after various tests, including adding fixed effects, replacing the CPU indicator, lagging by one period, applying two-way clustering robust standard errors, and using the instrumental variable (IV) method.

2 Theoretical analysis and hypotheses

The impact of CPU on firms' green innovation has received significant attention, but research remain inconclusive (Li et al., 2024; Sun et al., 2024; Bai et al., 2023; Niu et al., 2023; Huo et al., 2023). This paper argues that such divergences primarily stem from the different manifestations of CPU and the policy environment. Specifically, CPU is mainly reflected in two forms: first, the volatility and lack of continuity in climate policies, which often lead to a significant rise in sunk costs and an increase in default risks for firms; second, the deviation between the intensity of climate policy implementation by governments and expectations, although the policy itself retains a certain degree of continuity.

When climate policies are volatile and lack continuity, firms are forced to confront significantly higher sunk costs and rising default risks. This compels firms to adopt a more cautious approach in their investment decisions, particularly when these decisions involve green R&D projects, where this caution is even more pronounced (Bhattacharya et al., 2017; Sun et al., 2024). Moreover, based on real options theory, given the high costs and irreversibility associated with green innovation projects (Cui et al., 2023), decision-makers may be inclined to delay such projects until they can obtain sufficient information to mitigate potential risks and investment losses stemming from CPU (Gulen and Ion, 2016; Hoang, 2022; Sun et al., 2024).

However, when the intensity of climate policy implementation exceeds expectations and the policy maintains continuity, such uncertainty can instead present new opportunities for firms (Liu et al., 2017). According to growth options theory, green innovation can be viewed as a call option, where CPU exacerbates market risk, thereby driving industry restructuring. In this context, firms that invest early and succeed in green innovation projects are likely to achieve higher profit growth and secure more future investment opportunities, thus consolidating or regaining their market position (Aghion et al., 2005).

The overall effect of CPU on green innovation is complex, hindered in countries with fluctuating policies, in contrast with countries with relatively consistent climate goals, creating a more favorable market environment for green investment. As the latter is the case in China, we test the following claim:

Hypothesis: CPU positively affect corporate green innovation.

3 Research design

3.1 Sample and data

For our empirical analysis, we utilize a database comprising Chinese A-share listed companies from 2010 to 2022. We select 2010 as the starting year, as it coincides with the initial proposal by the Chinese government for a carbon emissions reduction target at the end of 2009. Following Cui et al. (2023), we process the original data as follows:

1. Excluding financial firms and firms marked with Special Treatment (ST), Special Treatment* (ST*), or Particular Transfer (PT)¹;
2. Eliminating observations with substantial missing data and remove samples with significant data biases, such as debt ratios less than 0 or greater than 1.
3. After this data cleaning process, we obtain data for 2,302 A-share listed companies and 20,566 observation samples. The statistical information on the Chinese national and provincial CPU index is sourced from [Ma et al. \(2023\)](#). Data on corporate green innovation is collected from CNRDS ². Additionally, all other variables are obtained from CSMAR database ³.

3.2 Variable descriptions

3.2.1 Green innovation

Following [Sun et al. \(2024\)](#), we measure corporate green innovation using the total number of green patent applications. The rationale for this approach is twofold:

1. Green patent data obtained from the National Intellectual Property Office demonstrates robust data accessibility and reliability.
2. Considering that the patent application process lasts from one to two years, and green patents impact the company within this period ([Li et al., 2016](#)), patent applications serve as a timelier indicator of green innovation compared to granted patents.

Furthermore, the number of green patents is a non-negative count variable, exhibiting a right-skewed distribution with a substantial concentration of observations at zero, making log transformation infeasible. A common approach involves using the natural logarithm of one plus the outcome to alleviate right-skewed distribution issues and retain observations with zero ([He et al., 2022](#); [Cui et al., 2023](#); [Tang et al., 2023](#); [Wang and Chu, 2024](#)). However, recent literature indicates that this approach produces estimates that lack a natural interpretation and can have an incorrect sign in expectation ([Cohn et al., 2022](#)); as such, we do not follow this approach.

3.2.2 Climate policy uncertainty (CPU)

The CPU index, initially developed by [Gavrilidis \(2021\)](#), serves as an instrument for measuring climate policy volatility and has been extensively utilized in existing literature ([Ren et al., 2022](#); [Sun et al., 2024](#); [Bai et al., 2023](#)). Following [Gavrilidis \(2021\)](#), some scholars have constructed the Chinese CPU index ([Xu et al., 2023](#); [Lee and Cho, 2023](#)). However, there are two main issues with these indices, 1) the CPU index is a country-level time-series indicator. Due to multi-collinearity, we cannot include time-fixed effects in the panel data regression model, which poses a challenge in removing the impact of other macro factors on firms ([Huang et al., 2022](#)). 2) the utilization of the CPU index assumes that all firms have the same perception of uncertainty. In reality, diverse regions within the same country experience disparate climate policies ([Nie et al., 2020](#)). Relying solely on the national CPU index impedes its ability to distinguish regional variations in the perception of policy uncertainty.

To address these issues, [Ma et al. \(2023\)](#) expanded the national CPU index to the provincial and municipal levels based on data from 1,755,826 articles published in six leading newspapers in China⁴

¹ST (Special Treatment) refers to a special treatment of the stock transactions of listed companies with abnormal financial conditions or other irregularities, serving as a warning of significant risk. *ST (Special Situations Treatment) indicates that the company has incurred losses for three consecutive years, warning of the risk of compulsory delisting. PT (Particular Transfer) refers to the suspension of listing for companies that have experienced losses for three consecutive years.

²Chinese Research Data Services Platform

³China Stock Market and Accounting Research

⁴The People's Daily, Guangming Daily, Economic Daily, Global Times, Science and Technology Daily, and China News Service.

for information pertaining to climate policies and uncertainties, combining manual auditing with the MacBERT model, a deep learning algorithm. Furthermore, inspired by the methodology employed by [Baker et al. \(2016\)](#), they standardized the data to establish the CPU Index. Therefore, we utilize the CPU index formulated by [Ma et al. \(2023\)](#). For our study, we use the natural logarithm of the provincial-level CPU index ($\ln CPU$).

3.2.3 Control variables

We incorporate a set of control variables based on existing literature. Specifically, the control variables are as follows:

1. Firm size ($\ln Size$): Measured as the natural logarithm of year-end total assets.
2. Business performance (ROA): Measured as the ratio of total net income to total assets.
3. Ownership concentration (Top_10): Measured as the shareholding ratio of the top ten shareholders.
4. Debt level (Lev): Measured as the ratio of total debts to total assets.
5. Firm age ($\ln Age$): Measured as the natural logarithm of the number of years the company has been listed plus one.

The descriptive statistics are shown in [Table 1](#).

Table 1: Descriptive statistics

Variable	Obs	Mean	Std. dev.	Min	Max
GP	22,226	4.628	28.497	0.000	1,166
$\ln CPU$	22,226	0.795	0.221	-0.116	1.381
$\ln size$	22,219	22.272	1.374	16.185	28.636
ROA	20,532	0.031	4.850	-687.864	73.367
Top10	22,223	0.589	0.155	0.013	1.000
Lev	22,218	0.417	0.202	0.010	1.000
$\ln age$	22,226	2.909	0.327	0.000	4.220

3.3 Baseline model

To identify the impact of CPU on corporate green innovation, we construct the following baseline regression model:

$$GP_{i,t} = \alpha_0 + \beta_0 \ln(CPU_{i,t}) + \sum \gamma_k Control_{i,k,t} + \vartheta_i + \varphi_t + \epsilon_{i,t} \quad (1)$$

where i, t represent firm and year, respectively. $GP_{i,t}$ is the total number of green patent applications. $\ln CPU_{i,t}$ is the natural logarithm of the provincial-level CPU index. The estimated coefficient of $\ln CPU_{i,t}$, denoted as β_0 , is our interest. The positive and statistically significant value of β_0 indicates that an increase in CPU promotes corporate green innovation. $Control_{i,k,t}$ represent a series of firm characteristic variables. ϑ_i and φ_t represent firm- and year- fixed effects, and $\epsilon_{i,t}$ represent an error term. Given that the green patent data is a non-negative count variable, we employ panel Poisson regression with fixed effects to estimate Equation 1, following [Cohn et al. \(2022\)](#).

4 Empirical results

4.1 Baseline results

The findings of CPU's influence on corporate green innovation are shown in [Table 2](#). Columns (1) control for firm- and year- fixed effects, while Columns (2) further control for a series of firm characteristic variables. The estimated coefficients of $\ln CPU$ are statistically significant at the 1% confidence level in

Columns (1) and (2). The results displayed in Table 1 indicate that CPU significantly promotes corporate green innovation. Interpreting the estimators in Columns (2), we find that the impact of CPU on corporate green innovation is approximately 0.467. After calculation, the incidence rate ratio is 1.596 ($e^{0.467}$), indicating that a 1-unit increase in CPU is associated with the number of green patent applications will become 1.596 times higher. The results indicate that, in the context of climate policies that are tightening yet maintaining a certain degree of continuity, enterprises tend to perceive this uncertainty as a call option. They actively promote green innovation within their companies by increasing investments in green research and development, thereby enhancing their market competitiveness.

Table 2: Baseline results

VARIABLES	GP (1)	GP (2)
lnCPU	0.548*** (0.183)	0.467*** (0.174)
Controls	NO	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	22,226	20,064
Pseudo R2	0.804	0.813

Notes: The standard error reported in parentheses is clustered at the firm level. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

To address omitted variable bias and reverse causality, we use city and year fixed effects, industry-year interaction fixed effects, and an IV regression (respectively), with the standard deviation of provincial temperature fluctuations as an IV for CPU. The standard deviation of provincial average temperature over the past five years (ATsd) is valid due to its strong correlation with CPU and its exogeneity from firm-level green innovation.⁵

Table 3 presents the results of the IV regression. The calculated coefficient of lnCPU in column (1) indicates a strong negative association between CPU and temperature fluctuation, significant at the 1% confidence level. The weak IV test shows that the Cragg-Donald Wald F statistics exceed 10, confirming the reliability of temperature fluctuation as an IV. Column (2) present the results of the second-stage regression, where the estimated coefficient of lnCPU is significantly positive at the 5% confidence level. These results are in line with the baseline findings.⁶

5 Conclusion and discussion

Using the provincial CPU index from a micro viewpoint, we empirically investigate the effect of CPU on corporate green innovation based on panel data for Chinese A-share listed enterprises from 2010 to 2022. Our results demonstrate that CPU has a favorable impact on corporate green innovation, and this finding holds up to robustness tests, including the addition of fixed effects, replacing the CPU indicator, lagging one period, using two-way clustering robust standard errors, and IV regression. Heterogeneity

⁵Greater temperature fluctuations increase the likelihood of more frequent adjustments in climate policies, such as CPU. Temperature fluctuation represents an objective climatic phenomenon that is minimally susceptible to influence from the green innovations of individual firms.

⁶In the online appendix, we do extensive robustness checks which confirm the reliability of our baseline results. First, incorporating industry- and province-fixed effects to account for potential unobservable variations over time yields a significantly positive coefficient for lnCPU. Second, substituting the national CPU index for the provincial index maintains a significant positive effect on corporate green innovation at the 1% level. Third, considering potential time lags in green innovation, a one-period lagged CPU variable still produces a significantly positive impact. Fourth, adjusting standard errors by clustering at both firm and year levels accounts for serial correlation, with results remaining robust. Fifth, addressing potential over-dispersion in green innovation counts using a panel negative binomial regression confirms the significance of lnCPU at the 10% level. Lastly, substituting green invention patent applications (GIP) for green patent applications as the explanatory variable continues to show a significantly positive effect.

Table 3: The results of IV regression

VARIABLES	First stage	Second stage
	(1)	(2)
lnCPU		64.504** (32.649)
ATsd	-0.016*** (0.002)	
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Cragg-Donald Wald F statistic		86.876
Observations	20,064	20,064

Notes: The standard error reported in parentheses is clustered at the firm level. ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

analyses show that CPU has a significant positive effect on green innovation in firms with high pollution levels, non-state-owned enterprises, higher green awareness among managers, lower risk-taking, and lower financing constraints.

This paper draws several policy implications. The positive impact of CPU on corporate green innovation suggests that the government should consider the uncertainty factor when designing climate policies and adopt flexible policy instruments to provide enterprises with more space and incentives for development. Additionally, the government should promote and facilitate green innovation activities by providing companies with comprehensive support in terms of finance and taxation, thereby aiding their transition to a green and low-carbon environment. ⁷

⁷Heterogeneity analysis suggests that the government should implement tailored support policies, such as enhancing green management training for firms with environmentally aware leadership, increasing financing support for non-state-owned enterprises to drive green innovation, and easing financing costs for small and micro-enterprises to improve access to financial resources.

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A Online Appendix

A.1 Robustness checks

The following robustness tests enhance the reliability and stability of the basic results.

First, we augment the model by incorporating the interaction term of industry- and province-fixed effects, as specified in Equation 1, to mitigate unobservable variations at the industry and province levels that evolve over time. The results displayed in columns (1) of Table 4 reveal that the estimated coefficient of $\ln\text{CPU}$ remain significantly positive.

Second, we substitute the national CPU index for the provincial CPU index and re-estimate the regression model. The results presented in columns (2) of Table 4 show that CPU still significantly and positively affects corporate green innovation at the 1% confidence level.

Third, recognizing that firms must undergo stages such as research and development, experimentation, and validation in green innovation processes that can be time-consuming, the impact of CPU on corporate green innovation may not be immediate. To address potential time lags, we adopt a one-period lag for the core explanatory variable, CPU. The estimated coefficient of $\ln\text{CPU}$ in column (3) of Table 4 is significantly positive at the 5% level.

Fourth, while standard errors are clustered at the firm level in all of the above regressions, our sample involves firm-year level observations, where residuals may exhibit serial correlation between different firms within the same year and between different years within the same firm. Therefore, we adjust standard errors by clustering at both the firm and year levels. The results shown in column (4) of Table 4 reveal that the estimated coefficient of $\ln\text{CPU}$ is positive and statistically significant.

Fifth, considering the possibility of over-dispersion in the number of corporate green innovation, we re-estimated using a panel negative binomial regression with fixed effects. The estimated coefficient of $\ln\text{CPU}$ in column (5) of Table 4 is 0.127 and significantly positive at the 10% level, which indicates to a certain extent that the problem of over-dispersion in the number of green patent applications has not interfered with our results.

Sixth, we use green invention patent applications (GIP) as an explanatory variable to replace green patent applications. The results displayed in columns (6) of Table 4 reveal that the estimated coefficient of $\ln\text{CPU}$ remain significantly positive at the 5% confidence level.

In summary, these robustness checks verify our baseline results.⁸

A.2 Heterogeneity analysis

Our findings reveal that CPU can positively affect corporate green innovation, but they do not indicate whether this impact varies under different conditions. Therefore, we further explore the heterogeneity of CPU's effect on corporate green innovation by examining factors such as pollution intensity, ownership type, managers' green concern, risk-taking behavior, and financing constraints.

A.2.1 Heterogeneity in corporate pollution intensity

During the process of climate policy adjustment, companies with varying pollution intensities may exhibit distinct inclinations towards green technology innovation, influenced by differences in environmental requirements, regulations, and motivations for green innovation. Therefore, we argue that CPU exerts a heterogeneous impact on green innovation among companies with different levels of pollution intensity. To examine this hypothesis, we categorize the sample into two subgroups based on whether the companies

⁸An online appendix contains heterogeneity analysis related to corporate pollution intensity, ownership type, managerial concerns, corporate risk-taking, and financial constraints.

Table 4: Robustness Checks

VARIABLES	GP	GP	GP	GP	GP	GIP
	(1)	(2)	(3)	(4)	(5)	(6)
lnCPU	0.479*** (0.173)	0.983*** (0.091)	0.499** (0.204)	0.467** (0.221)	0.127* (0.072)	0.559** (0.242)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Province Fe	Yes	No	No	No	No	No
Industry FE	Yes	No	No	No	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,059	20,290	19,087	20,064	20,064	17,346
Pseudo R2	0.815	0.808	0.814	0.813		0.817

Note: In column (3), lnCPU is calculated by the natural logarithm of the national CPU index. Since the national CPU index is a macro-level variable, we only control for a time trend term rather than year-fixed effects to mitigate the effects of time factors. In columns (4), lnCPU represents the one-period lag of lnCPU. Standard errors are clustered at two-way clustered at the firm and year levels in column (4). ***, **, and * indicate 1%, 5%, and 10% significance, respectively.

are identified as major polluters. Companies listed as major polluters are categorized as high pollution-intensity firms, while the rest are considered low pollution-intensity firms.

Panel A in Figure 1 displays the results of the heterogeneous impact of CPU on corporate green innovation, segmented by pollution intensity. The results reveal that the estimated coefficient of lnCPU is statistically significant at the 5% confidence level for high pollution-intensity firms. In contrast, the estimated coefficient is not statistically significant for low pollution-intensity firms.

These findings indicate that CPU significantly enhances corporate green innovation in high pollution-intensity firms but has no significant impact on low pollution-intensity firms. The rationale is that companies with high pollution intensity are more sensitive to climate policy adjustments due to the increased regulatory pressure and oversight. CPU could heighten the risk perception for these companies, prompting them to invest more in green technology innovation. Conversely, climate policy changes have a minimal effect on businesses with low emission intensity, resulting in less motivation for these firms to innovate in the green space.

A.2.2 Heterogeneity in ownership type

Companies with different ownership types may exhibit varying responses and strategies to cope with adjustments in climate policies, influenced by factors such as internal management mechanisms, market positioning, and sensitivity to policies. Therefore, we hypothesize that CPU exerts a heterogeneous impact on green innovation among companies with diverse ownership structures. To validate this assertion, we categorize companies into state-owned enterprises (SOEs) and non-state-owned enterprises (non-SOEs) based on their ownership type.

We run the Eq.1 for the two subgroups and report the results in Panel B in Figure 1. The results reveal that the estimated coefficients of lnCPU are statistically insignificant for non-SOEs but significantly positive for SOEs, indicating that CPU positively influences corporate green innovation in non-SOEs, while it has no significant impact on SOEs.

A possible explanation is that non-SOEs, with their market flexibility and stronger profit incentives, tend to invest more in green innovation to respond to market needs and enhance competitiveness. On the other hand, SOEs typically have clear corporate missions and strategic goals that prioritize policy stability and consistency. Even when facing CPU, state-owned enterprises may rely on government support and guidance, leading to comparatively lower investments in green innovation.

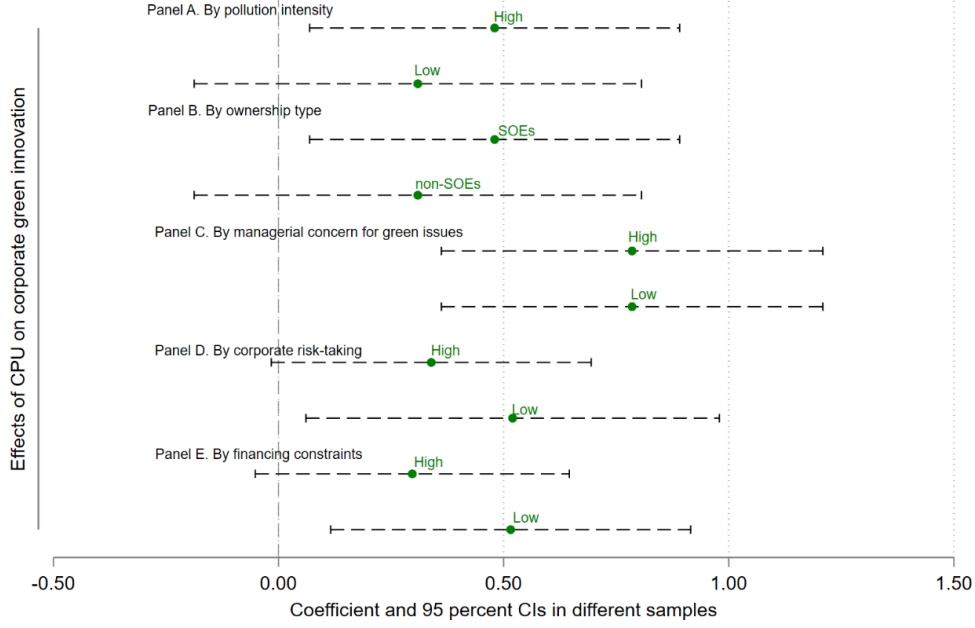


Figure 1: Heterogeneous effects of CPU on corporate green innovation

A.2.3 Heterogeneity in managerial concern for green issues

Green innovation is significantly influenced by senior management, which drives strategic decision-making and resource allocation within the organization (Wei et al., 2022). Managerial concern for green issues substantially affects their perception of CPU and their response behavior, influencing the degree to which enterprises invest in and pursue green innovation. Therefore, we argue that CPU exerts a heterogeneous impact on green innovation depending on managerial concern for green issues. To test this hypothesis, we divide the companies into two subgroups based on the median value of the frequency of green-related words. Following Duriau et al. (2007), we identify green-related keywords from the annual reports of listed companies and use word frequency statistics to measure managerial concern for environmental issues. Furthermore, we categorize companies into two subgroups: high managerial concern and low managerial concern for environmental issues.

In panel C in Figure 1, we run the regression for companies with a high level of managerial concern for green issues. The results reveal that the estimated coefficient of $\ln CPU$ is significantly positive at the 1% confidence level. Additionally, we run the regression for companies with a low level of managerial concern for green issues. The results demonstrate that the estimated coefficient of $\ln CPU$ is statistically insignificant.

These findings suggest that CPU has a favorable effect on green innovation among companies with a high level of managerial concern for green issues, but it has no significant impact on those with a low level of managerial concern for green issues. One possible explanation is that managers with a high level of concern for environmental issues are more inclined to recognize opportunities in adjusting climate policies, leading to enhanced company profitability through green innovation. Conversely, managers with lower environmental awareness often do not emphasize climate policy adjustments, resulting in less impact on green innovation.

A.2.4 Heterogeneity in corporate risk-taking

The level of corporate risk-taking may influence its performance and conduct regarding green practices (Yan et al., 2021). Therefore, we hypothesize that CPU exerts a heterogeneous impact on green innovation

among companies with different levels of risk-taking. To test this hypothesis, we split organizations into two groups based on the median value of corporate risk-taking.

Following [John et al. \(2008\)](#), we calculate the standard deviation of the adjusted ROA over the last three years to gauge corporate risk-taking. Companies with a risk-taking index higher than the median value are defined as high risk-taking companies, while those below the median are defined as low risk-taking companies.

Panel D in Figure 1 reports the impact of CPU on green innovation among enterprises with varying levels of corporate risk-taking. The estimated coefficient of $\ln CPU$ can be found to be 0.339 for high risk-taking firms, which is statistically significant at the 10 per cent level, while for low risk-taking firms, the estimated coefficient on $\ln CPU$ is 0.520, which is statistically significant at the 5 per cent level.

These results indicate that CPU exerts a more pronounced positive impact on companies with low risk-taking compared to those with high risk-taking. One possible explanation is that companies with low risk-taking might increase their green R&D investment in response to the pressures and risks associated with CPU. Conversely, companies with strong risk-taking capabilities exhibit notable adaptability, allowing them to adeptly handle policy shifts and maintain their green innovation efforts.

A.2.5 Heterogeneity in financing constraints

Corporate green innovation is influenced by financial constraints. As CPU increases, companies with varying levels of financing constraints may adopt different green innovation strategies. Following [Livdan et al. \(2009\)](#), We measure corporate financing constraints in the index constructed by [Whited and Wu \(2006\)](#) (WW index), where a larger WW index indicates higher financing constraints for the company. We then divide the sample into two subgroups based on the WW index, categorizing companies with a WW index exceeding the median value as those facing high financing constraints and those with a WW index below the median as those with low financing constraints.

Panel E in Figure 1 reports the impact of CPU on green innovation among enterprises with varying levels of cor financing constraints. For companies confronting high financing constraints, the results indicate that the estimated coefficient of $\ln CPU$ is 0.297, which exhibits statistical significance at the 10% confidence level. Conversely, for companies facing low financing constraints, the estimated coefficient of $\ln CPU$ stands is 0.516, demonstrating statistical significance at the 5% level.

These findings suggest that companies with low financing constraints benefit more from CPU than those with high financing constraints. The rationale is that companies with low financing constraints usually have relatively ample funds, enabling them to dynamically allocate resources for green technology research and innovation in response to climate policy adjustments. In contrast, companies with high financing constraints may tend to be more conservative and cautious due to financial pressures, potentially leading to limited investment in green innovation.

Furthermore, the heterogeneity analyses indicates that the government should design differentiated support policies for different types of enterprises. For example, for companies with strong green awareness among their managers, relevant green management training and guidance can be strengthened to enhance their cognitive and practical capabilities in green innovation. Provide more financing support to non-state-owned enterprises to leverage their leading role in green technology innovation. Additionally, policies to support the financing of small and micro-enterprises can be adopted to help companies with modest funding limitations lower their financing costs and facilitate access to financial support.